

## PRACTICE SHEET

1.  $\lim_{x \rightarrow 0} \frac{\sin |x|}{x}$  equal to ?  
(a) 1 (b) -1  
(c)  $\infty$  (d) Limit D.N.E.
2.  $\lim_{x \rightarrow 0} \frac{x \sin 5x}{\sin^2 4x}$  equal to ?  
(a) 0 (b) 5/4  
(c) 5/16 (d) 25/4
3.  $\lim_{x \rightarrow \infty} \left( \frac{x-2}{x+2} \right)^{x+2}$  equal to ?  
(a) 0 (b)  $e^4$   
(c)  $e^{-2}$  (d)  $e^{-4}$
4.  $\lim_{x \rightarrow \infty} \frac{\sin x}{x}$  equal to ?  
(a) 1 (b) 0  
(c)  $\infty$  (d) -1
5.  $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$  equal to ?  
(a)  $\log \left( \frac{a}{b} \right)$  (b)  $\log \left( \frac{b}{a} \right)$   
(c)  $ab$  (d)  $\log(ab)$
6.  $\lim_{x \rightarrow \infty} \sec \theta - \tan \theta$  is equal to  
(a) 0 (b) 1  
(c) -1 (d) 2
7.  $\lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1}$  equal to ?  
(a)  $\log 2$  (b)  $2 \log 2$   
(c)  $\frac{1}{2} \log 2$  (d) 0
8.  $\lim_{x \rightarrow \infty} x \cos \left( \frac{\pi}{4x} \right) \sin \left( \frac{\pi}{4x} \right)$  equal to ?  
(a)  $\frac{\pi}{4}$  (b)  $\frac{\pi}{3}$   
(c)  $\pi$  (d) 0
9.  $\lim_{x \rightarrow \infty} \left[ x - \sqrt{x^2 - x} \right]$  equal to ?  
(a) 1/2 (b) 1  
(c) -1/2 (d) 0
10.  $\lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x^2 \sin x}$  equal to ?  
(a) 1/2 (b) -1/2  
(c) 1/3 (d) -1/3
11. What is  $\lim_{x \rightarrow \infty} \left( \frac{x}{3+x} \right)^{3x}$  equal to?  
(a)  $e$  (b)  $e^3$   
(c)  $e^{-9}$  (d)  $e^9$
12. What is  $\lim_{x \rightarrow 0} \frac{\sin^2 ax}{bx}$  (a, b are constants) equal to?  
(a) 0 (b) a  
(c)  $\frac{a^2}{b}$  (d) Does not exist
13.  $\lim_{x \rightarrow 0} e^{-1/x}$  is equal to:  
(a) 0 (b)  $\infty$   
(c)  $e$  (d) Does not exist
14. If  $f(x) = \frac{1}{1-|1-x|}$ , then what is  $\lim_{x \rightarrow 1} f(x)$  equal to?  
(a) 0 (b)  $\infty$   
(c) 1 (d) -1
15. What is the value of  $\lim_{x \rightarrow \alpha} \frac{\sqrt{\alpha+2x} - \sqrt{3x}}{\sqrt{3\alpha+x} - 2\sqrt{x}}$ ?  
(a)  $\frac{2}{\sqrt{3}}$  (b)  $\frac{1}{(3\sqrt{3})}$   
(c)  $\frac{2}{(3\sqrt{3})}$  (d)  $\frac{1}{\sqrt{3}}$
16. What is  $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$ ?  
(a)  $\log \left( \frac{a}{b} \right)$  (b)  $\log \frac{b}{a}$   
(c)  $ab$  (d)  $\log(ab)$
17. What is the value of  $\lim_{x \rightarrow \infty} \left( \frac{x+6}{x+1} \right)^{x+4}$ ?  
(a)  $e$  (b)  $e^2$   
(c)  $e^4$  (d)  $e^5$
18. What is the value of  $\lim_{x \rightarrow 1} \frac{(x-1)^2}{|x-1|}$ ?  
(a) 0 (b) 1  
(c) -1 (d) Does not exist

### ANSWER KEY

|     |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |    |   |     |   |
|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|----|---|-----|---|
| 1.  | d | 2.  | c | 3.  | d | 4.  | b | 5.  | a | 6.  | a | 7.  | b | 8.  | a | 9. | a | 10. | d |
| 11. | c | 12. | a | 13. | d | 14. | c | 15. | c | 16. | a | 17. | d | 18. | a |    |   |     |   |

## Solutions

**Sol.1. (d)**

$\lim_{x \rightarrow 0} \frac{\sin |x|}{x}$ , L.H.L is limit when  $x < 0$

$$\text{LHL} = \lim_{x \rightarrow 0} \frac{\sin(x)}{x} = - \lim_{x \rightarrow 0} \frac{\sin x}{x} = -1$$

$$\text{RHL} = \lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

So, LHL  $\neq$  RHL

Hence, limit it does not exist

**Sol.2. (c)**

$$\lim_{x \rightarrow 0} \frac{x \sin 5x}{\sin^2 4x}$$

[Multiply denominator and number with x]

$$\lim_{x \rightarrow 0} \frac{x^2 \sin 5x}{x \sin^2 4x} = \lim_{x \rightarrow 0} \frac{\sin 5x}{x \rightarrow 0} \cdot \frac{x^2}{\sin^2 4x}$$

Rearranging to bring a standard form, we get,

$$\lim_{x \rightarrow 0} \frac{5 \sin 5x}{5x} \cdot \frac{(4x)^2}{16 \sin^2 4x} = \frac{5}{16} \left( \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \right) \cdot \frac{1}{\lim_{x \rightarrow 0} \left( \frac{\sin 4x}{4x} \right)^2} = \frac{5}{16}$$

**Sol.3. (d)**

$$\lim_{x \rightarrow 0} \left( \frac{x-2}{x+2} \right)^{x+2}$$

can be written as

$$\lim_{x \rightarrow 0} \left( \frac{x+2-4}{x+2} \right)^{x+2}$$

$$\text{or } \lim_{x \rightarrow 0} \left\{ 1 - \frac{4}{x+2} \right\}^{x+2}$$

putting  $x+2=t$

when  $x \rightarrow \infty$ ,  $t \rightarrow \infty$

$$\text{so, } \lim_{x \rightarrow 0} \left\{ 1 - \frac{4}{t} \right\}^t$$

$$\text{or } \lim_{x \rightarrow 0} \left\{ 1 - \frac{4}{t} \right\}^{\frac{t}{4} \times 4}$$

$$\text{or, } \lim_{x \rightarrow 0} \left\{ \left( 1 - \frac{4}{t} \right)^{\frac{t}{4}} \right\}^4 = (e^{-1})^4 = e^{-4}$$

**Sol.4. (b)**

$$\text{Given limit is : } \lim_{x \rightarrow 0} \frac{\sin}{x}$$

Let,  $x=1/h$  and as  $x \rightarrow \infty$ ,  $h \rightarrow 0$

$$\therefore \lim_{x \rightarrow 0} \frac{\sin x}{x} \text{ change to : } \lim_{h \rightarrow 0} h \sin \frac{1}{h}$$

$=0$  (value of  $\sin 1/h$  is finite as it lies between  $-1$  and  $1$ )

**Sol.5. (a)**

$$\text{Let } L = \lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$$

This is of  $0/0$  form so, L' hospital's rule is applicable

$$\Rightarrow L = \lim_{x \rightarrow 0} \frac{a^x \log a - b^x \log b}{1}$$

(by L' Hospital's rule)

$$= \log a - \log b = \log a/b$$

**Sol. 6. (a)**

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \sec \theta - \tan \theta = \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin \theta}{\cos \theta}$$

apply L hospital

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{-\cos \theta}{-\sin \theta} = 0$$

**Sol. 7. (b)**

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1}$$

apply L hospital

$$\lim_{x \rightarrow 0} \frac{2^x \log 2}{\frac{1}{2\sqrt{1+x}}} = 2 \log 2$$

**Sol. 8. (a)**

$$\lim_{x \rightarrow \infty} x \cos \left( \frac{\pi}{4x} \right) \sin \left( \frac{\pi}{4x} \right)$$

$$\lim_{x \rightarrow \infty} \frac{x}{2} \sin \left( \frac{\pi}{2x} \right) = \lim_{x \rightarrow \infty} \frac{\sin \left( \frac{\pi}{2x} \right)}{\frac{2}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{\sin \left( \frac{\pi}{2x} \right)}{\frac{\pi}{2x}} \cdot \frac{\pi}{4} = \frac{\pi}{4}$$

**Sol. 9. (a)**

$$\lim_{x \rightarrow \infty} \left[ x - \sqrt{x^2 - x} \right]$$

$$= \lim_{x \rightarrow \infty} \left[ x - \sqrt{x^2 - x} \right] \times \frac{x + \sqrt{x^2 - x}}{x + \sqrt{x^2 - x}}$$

$$= \lim_{x \rightarrow \infty} \frac{x}{x + \sqrt{x^2 - x}} = \frac{1}{2}$$

**Sol. 10. (d)**

$$\lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x^2 \sin x} \left( \frac{0}{0} \text{ form} \right)$$

apply L hospital

$$\lim_{x \rightarrow 0} \frac{\cos x - x \sin x - \cos x}{x^2 \cos x + 2x \sin x} = \lim_{x \rightarrow 0} \frac{-\sin x}{x \cos x + 2 \sin x}$$

again apply L hospital

$$= \lim_{x \rightarrow 0} \frac{-\cos x}{\cos x - x \sin x + 2 \cos x} = -\frac{1}{3}$$

**Sol. 11. (c)**

$$\lim_{x \rightarrow \infty} \left( \frac{x}{3+x} \right)^{3x}$$

$$= \lim_{x \rightarrow \infty} \left( \frac{x+3-3}{3+x} \right)^{3x} = \lim_{x \rightarrow \infty} \left( 1 + \frac{-3}{3+x} \right)^{3x} = e^{-9}$$

**Sol. 12. (a)**

$$\lim_{x \rightarrow 0} \frac{\sin^2 ax}{bx}$$

$$\lim_{x \rightarrow 0} \frac{\sin^2 ax}{a^2 x^2} \times \frac{a^2 x}{b} = 1 \times \frac{a^2(0)}{b} = 0$$

**Sol. 13. (d)**

$$\lim_{x \rightarrow 0} e^{-1/x}$$

L.H.L.

$$\lim_{h \rightarrow 0} e^{-1/(0-h)} = \lim_{h \rightarrow 0} e^{1/h} = e^{\infty} = \infty$$

R.H.L.

$$\lim_{h \rightarrow 0} e^{-1/(0+h)} = e^{-\infty} = 0$$

L.H.L.  $\neq$  R.H.L so limit does not exist.

**Sol. 14. (c)**

$$\lim_{x \rightarrow 1} \frac{1}{1 - |1-x|} = 1$$

**Sol. 15. (c)**

$$\lim_{x \rightarrow \infty} \frac{\sqrt{\alpha+2x} - \sqrt{3x}}{\sqrt{3\alpha+x} - 2\sqrt{x}}$$

rationalize the numerator and denominator

$$\lim_{x \rightarrow \infty} \frac{\sqrt{\alpha+2x} - \sqrt{3x}}{\sqrt{3\alpha+x} - 2\sqrt{x}} \times \frac{\sqrt{3\alpha+x} + 2\sqrt{x}}{\sqrt{3\alpha+x} + 2\sqrt{x}} \times \frac{\sqrt{\alpha+2x} + \sqrt{3x}}{\sqrt{\alpha+2x} + \sqrt{3x}}$$

$$= \lim_{x \rightarrow \infty} \frac{\alpha+2x-3x}{3\alpha+x-4x} \times \frac{\sqrt{3\alpha+x} + 2\sqrt{x}}{\sqrt{\alpha+2x} + \sqrt{3x}}$$

$$\lim_{x \rightarrow \infty} \frac{\alpha-x}{3(\alpha-x)} \times \frac{\sqrt{3\alpha+x} + 2\sqrt{x}}{\sqrt{\alpha+2x} + \sqrt{3x}} = \frac{4}{6\sqrt{3}} = \frac{2}{3\sqrt{3}}$$

**Sol. 16. (a)**

$$\lim_{x \rightarrow 0} \frac{a^x - b^x}{x} ?$$

apply L. Hospital rule

$$\lim_{x \rightarrow 0} \frac{a^x \log a - b^x \log b}{1} = \log a - \log b = \log \left( \frac{a}{b} \right)$$

**Sol. 17. (d)**

$$\lim_{x \rightarrow \infty} \left( \frac{x+6}{x+1} \right)^{x+4}$$

$$\lim_{x \rightarrow \infty} \left( \frac{x+1+5}{x+1} \right)^{x+4} = \lim_{x \rightarrow \infty} \left( 1 + \frac{5}{x+1} \right)^{x+4}$$

by standard limit

$$= e^5$$

**Sol. 18. (a)**

$$\lim_{x \rightarrow 1} \frac{(x-1)^2}{|x-1|} ?$$

$$\lim_{x \rightarrow 1} \frac{|x-1|^2}{|x-1|} = \lim_{x \rightarrow 1} |x-1| = 0$$

## NDA PYQ

1. What is  $\lim_{x \rightarrow \infty} (\sqrt{a^2 x^2 + ax + 1} - \sqrt{a^2 x^2 + 1})$  equal to?  
(a) 1/2 (b) 1  
(c) 2 (d) 0  
[NDA (I) - 2011]
2. What is the value of  $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$ ?  
(a) 0 (b) 1  
(c) 1/2 (d) limit does not exist  
[NDA (I) - 2012]
3. What is the value of  $\lim_{x \rightarrow -2} \left(\frac{x+2}{x^3+8}\right)$ ?  
(a) 1/4 (b) -1/4  
(c) 1/12 (d) -1/12  
[NDA (I) - 2012]
4. What is the value of  $\lim_{x \rightarrow 0} \frac{2(1-\cos x)}{x^2}$ ?  
(a) 0 (b) 1/2  
(c) 1/4 (d) 1  
[NDA (II) - 2012]
5. What is the value of  $\lim_{x \rightarrow 2} \frac{x-2}{x^2-4}$ ?  
(a) 0 (b) 1/4  
(c) 1/2 (d) 1  
[NDA (II) - 2012]
6. Consider the following statements:  
I.  $\lim_{x \rightarrow 0} \frac{1}{x}$  exists.  
II.  $\lim_{x \rightarrow 0} e^{1/x}$  does not exist  
Which of the above statements is/are correct?  
(a) Only I (b) Only II  
(c) Both I and II (d) Neither I nor II  
[NDA (II) - 2012]
7. What is the value of  $\lim_{x \rightarrow 0} \frac{\sqrt{1+x}-1}{x}$ ?  
(a) 0 (b) 1/2  
(c) 1 (d) -1/2  
[NDA (II) - 2012]
8. Consider the following statements:  
I.  $\lim_{x \rightarrow 0} \sin \frac{1}{x}$  does not exist  
II.  $\lim_{x \rightarrow 0} x \sin \frac{1}{x}$  exists  
Which of the above statements is/are correct?  
(a) Only I (b) Only II  
(c) Both I and II (d) Neither I nor II  
[NDA (I) - 2013]
9. Which is  $\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x}$  equal to?  
(a) 0 (b) 1  
(c) -1 (d) 1/2  
[NDA (I) - 2013]
10. What is the value of  $\lim_{x \rightarrow 0} \frac{1-\sqrt{1+x}}{x}$ ?  
(a) 0 (b) 1  
(c) -1 (d) -1/2  
[NDA (I) - 2013]
11. What is  $\lim_{x \rightarrow 2} \frac{2-x}{x^3-8}$  equal to?  
(a) 1/8 (b) -1/8  
(c) 1/12 (d) -1/12  
[NDA (II) - 2013]
12. What is  $\lim_{x \rightarrow 0} \frac{1-\cos x}{x}$  equal to?  
(a) 0 (b) 1/2  
(c) 1 (d) 2  
[NDA (II) - 2013]
13. What is  $\lim_{x \rightarrow 0} \frac{\cos x}{\pi-x}$  equal to?  
(a) 0 (b)  $\pi$   
(c)  $1/\pi$  (d) 1  
[NDA (II) - 2013]
14. What is  $\lim_{x \rightarrow 0} \frac{\sin 2x + 4x}{2x + \sin 4x}$  equal to?  
(a) 0 (b) 1/2  
(c) 1 (d) 2  
[NDA (II) - 2013]
15.  $f(9) = 9$  and  $f'(9) = 4$ , then what is  $\lim_{x \rightarrow 9} \frac{\sqrt{f(x)}-3}{\sqrt{x}-3}$  equal to?  
(a) 36 (b) 9  
(c) 4 (d) None of these  
[NDA (I) - 2014]
16. What is  $\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x}$  equal to?  
(a) 0 (b) 1  
(c) n (d) n-1  
[NDA (I) - 2014]
17. What is  $\lim_{x \rightarrow 0} \frac{x}{\sqrt{1-\cos x}}$  equal to?  
(a)  $\sqrt{2}$  (b)  $-\sqrt{2}$   
(c)  $\frac{1}{\sqrt{2}}$  (d) Does not exist  
[NDA (I) - 2014]
18. What is  $\lim_{x \rightarrow 0} \frac{\log_5(1+x)}{x}$  equal to?  
(a) 1 (b)  $\log_5 e$   
(c)  $\log_e 5$  (d) 5  
[NDA (II) - 2014]
19. What is  $\lim_{x \rightarrow 0} \frac{5^x - 1}{x}$  equal to?  
(a)  $\log_e 5$  (b)  $\log_5 e$   
(c) 5 (d) 1  
[NDA (II) - 2014]

20. What is  $\lim_{n \rightarrow \infty} \frac{1+2+3+\dots+n}{1^2+2^2+3^2+\dots+n^2}$  equal to?
- (a) 5 (b) 2  
(c) 1 (d) 0

[NDA (II) - 2014]

21. If  $G(x) = \sqrt{25-x^2}$ , then what is  $\lim_{x \rightarrow 1} \frac{G(x)-G(1)}{x-1}$  equal to?
- (a)  $-\frac{1}{2\sqrt{6}}$  (b)  $\frac{1}{5}$   
(c)  $-\frac{1}{6}$  (d)  $\frac{1}{\sqrt{6}}$

[NDA (I) - 2015]

Directions (for next two): Given that

$$\lim_{x \rightarrow \infty} \left( \frac{2+x^2}{1+x} - Ax - B \right) = 3$$

22. What is the value of A?
- (a) -1 (b) 1  
(c) 2 (d) 3
23. What is the value of B?
- (a) -1 (b) 3  
(c) -4 (d) -3

[NDA (I) - 2015]

24. If  $f(x) = \sqrt{25-x^2}$ , then what is  $\lim_{x \rightarrow 1} \frac{f(x)-f(1)}{x-1}$  equal to?
- (a)  $\frac{1}{5}$  (b)  $\frac{1}{24}$   
(c)  $\sqrt{24}$  (d)  $-\frac{1}{\sqrt{24}}$

[NDA (II) - 2015]

25. If  $f(x) = \frac{\sin(e^{x-2}-1)}{\ln(x-1)}$ , then  $\lim_{x \rightarrow 2} f(x)$  is equal to?
- (a) -2 (b) -1  
(c) 0 (d) 1

[NDA (II) - 2015]

For the two (2) items that follow:

$$f(x) = \frac{a^{[x]+x} - 1}{[x] + x}$$

where  $[.]$  denotes the greatest integer function.

26. What is  $\lim_{x \rightarrow 0^+} f(x)$  equal to?
- (a) 1 (b)  $\ln a$   
(c)  $1 - a^{-1}$  (d) Limit does not exist
27. What is  $\lim_{x \rightarrow 0^-} f(x)$  equal to?
- (a) 0 (b)  $\ln a$   
(c)  $1 - a^{-1}$  (d) Limit does not exist
28. What is  $\lim_{x \rightarrow 0} e^{\frac{1}{x^2}}$  equal to?
- (a) 0 (b) 1  
(c) -1 (d) Limit does not exist

[NDA (I) - 2016]

[NDA (I) - 2016]

[NDA (I) - 2016]

29. If  $\lim_{x \rightarrow 0} \phi(x) = a^2$ , where  $a \neq 0$ , then what is  $\lim_{x \rightarrow 0} \phi\left(\frac{x}{a}\right)$  equal to?

- (a)  $a^2$  (b)  $a^{-2}$   
(c)  $-a^2$  (d)  $-a$

[NDA (I) - 2016]

30. What is  $\lim_{x \rightarrow 0} \frac{e^x - (1+x)}{x^2}$  equal to?
- (a) 0 (b)  $\frac{1}{2}$   
(c) 1 (d) 2

[NDA (I) - 2017]

31. If  $F(x) = \sqrt{9-x^2}$ , then what is  $\lim_{x \rightarrow 1} \frac{f(x)-F(1)}{x-1}$  equal to:
- (a)  $-\frac{1}{4\sqrt{2}}$  (b)  $\frac{1}{8}$   
(c)  $-\frac{1}{2\sqrt{2}}$  (d)  $\frac{1}{2\sqrt{2}}$

[NDA (I) - 2017]

32. Consider the following statements:
1. If  $\lim_{x \rightarrow a} f(x)$  and  $\lim_{x \rightarrow a} g(x)$  both exists, then  $\lim_{x \rightarrow a} \{f(x)g(x)\}$  exists.
2. If  $\lim_{x \rightarrow a} \{f(x)g(x)\}$  exists, then both  $\lim_{x \rightarrow a} f(x)$  and  $\lim_{x \rightarrow a} g(x)$  must exists.
- Which of the above statements is /are correct?
- (a) Only 1 (b) Only 2  
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2017]

33. If  $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{x} = l$  and  $\lim_{x \rightarrow \infty} \frac{\cos x}{x} = m$ , then which one of the following is correct?
- (a)  $l=1, m=1$  (b)  $l=\frac{2}{\pi}, m=-\infty$   
(c)  $l=\frac{2}{\pi}, m=0$  (d)  $l=1, m=\infty$

[NDA (II) - 2017]

34. What is  $\lim_{h \rightarrow 0} \frac{\sqrt{2x+3h} - \sqrt{2x}}{2h}$  equal to?
- (a)  $\frac{1}{2\sqrt{2x}}$  (b)  $\frac{3}{\sqrt{2x}}$   
(c)  $\frac{3}{2\sqrt{2x}}$  (d)  $\frac{3}{4\sqrt{2x}}$

[NDA (I) - 2018]

35. What is  $\lim_{x \rightarrow 0} \frac{\tan x}{\sin 2x}$  equal to?
- (a)  $1/2$  (b) 1  
(c) 2 (d) Limit does not exist

[NDA (I) - 2018]

36. What is  $\lim_{x \rightarrow \frac{\pi}{6}} \frac{2\sin^2 x + \sin x - 1}{2\sin^2 x - 3\sin x + 1}$  to?
- (a)  $-\frac{1}{2}$  (b)  $-\frac{1}{3}$   
(c) -2 (d) -3

[NDA (II) - 2018]

37. What is  $\lim_{\theta \rightarrow 0} \frac{\sqrt{1-\cos \theta}}{\theta}$  equal to?

(a)  $\sqrt{2}$   
(c)  $\frac{1}{\sqrt{2}}$

(b)  $2\sqrt{2}$   
(d)  $-\frac{1}{2\sqrt{2}}$

[NDA (II) - 2018]

38. If  $f(x) = \sqrt{25-x^2}$ , then what is  $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x-1}$  equal to:

(a)  $-\frac{1}{\sqrt{24}}$   
(c)  $-\frac{1}{4\sqrt{3}}$

(b)  $\frac{1}{\sqrt{24}}$   
(d)  $\frac{1}{4\sqrt{3}}$

[NDA (II) - 2018]

39.  $\lim_{x \rightarrow 0} \frac{1 - \cos^3 4x}{x^2}$  equal to?

(a) 0  
(c) 24

(b) 12  
(d) 36

[NDA (I) - 2019]

40. What is the value of  $\lim_{x \rightarrow 0} \frac{\sin x^0}{\tan 3x^0}$ ?

(a) 1/4  
(c) 1/2

(b) 1/3  
(d) 1

[NDA (II) - 2019]

41. What is  $\lim_{x \rightarrow 0} \frac{3^x + 3^{-x} - 2}{x}$  equal to?

(a) 0  
(c) 1

(b) -1  
(d) limit does not exist

[NDA 2020]

42. What is  $\lim_{x \rightarrow 1} \frac{x + x^2 + x^3 - 3}{x-1}$  equal to?

(a) 1  
(c) 3

(b) 2  
(d) 6

[NDA 2020]

43. If  $\lim_{x \rightarrow 1} \frac{x^4 - 1}{x-1} = \lim_{x \rightarrow k} \frac{x^3 - k^3}{x^2 - k^2}$ , where  $k \neq 0$ , then what is the value of  $k$ ?

(a) 2/3  
(c) 8/3

(b) 4/3  
(d) 4

[NDA 2020]

44. What is  $\lim_{x \rightarrow 0} \frac{\sin x \log(1-x)}{x^2}$  equal to?

(a) -1  
(c) -e

(b) 0  
(d) -1/e

[NDA 2020]

45. If a differentiable function  $f(x)$  satisfies

$\lim_{x \rightarrow -1} \frac{f(x)}{x^2 - 1} = -\frac{3}{2}$  then what is the value of  $\lim_{x \rightarrow -1} f(x)$ ?

(a)  $-\frac{3}{2}$   
(c) 0

(b) -1  
(d) 1

[NDA (I) - 2021]

46.  $\lim_{x \rightarrow a} \frac{a^x - x^a}{x^a - a^a} = -1$

then what is the value of  $a$ ?

(a) -1  
(c) 1

(b) 0  
(d) 2

[NDA (I) - 2021]

47. What is  $\lim_{x \rightarrow -1} \frac{x^3 + x^2}{x^2 + 3x + 2}$  equal to?

(a) 0  
(c) 2

(b) 1  
(d) 3

[NDA (I) - 2021]

48. What is  $\lim_{n \rightarrow \infty} \frac{a^n + b^n}{a^n - b^n}$  where  $a > b > 1$ , equal to?

(a) -1  
(c) 1

(b) 0  
(d) Limits does not exist

[NDA (II) - 2021]

49. Let  $f(x) = \begin{cases} 1 + \frac{x}{2k}, & 0 < x < 2 \\ kx, & 2 \leq x < 4 \end{cases}$

If  $\lim_{x \rightarrow 2} f(x)$  exists, then what is the value of  $k$ ?

(a) -2  
(c) 0

(b) -1  
(d) 1

[NDA (II) - 2021]

50. If  $f(x) = \frac{[x]}{|x|}$ ,  $x \neq 0$ , where  $[.]$  denotes the greatest integer function, then what is the right-hand limit of  $f(x)$  at  $x = 1$ ?

(a) -1  
(b) 0  
(c) 1

(d) Right-hand limit of  $f(x)$  at  $x = 1$  does not exist

[NDA (II) - 2021]

51. What is  $\lim_{x \rightarrow 0} x^3 (\operatorname{cosec})^2$  equal to?

(a) 0  
(c) 1

(b)  $\frac{1}{2}$   
(d) Limit does not exist

[NDA (I) - 2022]

52. What is  $\lim_{x \rightarrow 1} \frac{x^3 - 1}{\sqrt{x} - 1}$  equal to?

(a) 0  
(c) 6

(b) 3  
(d) Limit does not exist

[NDA (I) - 2022]

53. What is  $\lim_{x \rightarrow 0} \frac{x}{\sqrt{1 - \cos 4x}}$  equal to?

(a)  $\frac{1}{2\sqrt{2}}$   
(c)  $\sqrt{2}$

(b)  $-\frac{1}{2\sqrt{2}}$   
(d) Limit does not exist

[NDA 2022 (II)]

54. What is  $\lim_{x \rightarrow \frac{\pi}{2}} \frac{4x - 2\pi}{\cos x}$  equal to?

(a) -4  
(c) 2

(b) -2  
(d) 4

[NDA 2022 (II)]

55. If  $f(x) = \frac{x^2 + x + |x|}{x}$ , then what is  $\lim_{x \rightarrow 0} f(x)$  equal to?

(a) 0  
(c) 2

(b) 1  
(d)  $\lim_{x \rightarrow 0} f(x)$  does not exist

[NDA 2022 (II)]

56. What is  $\lim_{h \rightarrow 0} \frac{\sin^2(x+h) - \sin^2 x}{h}$  equal to?

(a)  $\sin^2 x$   
(c)  $\sin 2x$

(b)  $\cos^2 x$   
(d)  $\cos 2x$

[NDA 2022 (II)]

57. What is  $\lim_{x \rightarrow 5} \frac{5-x}{|x-5|}$  equal to?

- (a) -1 (b) 0  
(c) 1 (d) Limit does not exist

[NDA – 2023 (1)]

58. What is  $\lim_{x \rightarrow 1} \frac{x^9 - 1}{|x^3 - 1|}$  equal to?

- (a) -1 (b) -3  
(c) 3 (d) Limit does not exist

[NDA – 2023 (1)]

Consider the following for the next three (03) items that follow:

$$\text{Let } f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2 \sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$$

59. What is  $f(0)$  equal to?

- (a) -1 (b) 0  
(c) 1 (d) 2

[NDA – 2023 (1)]

60. What is  $\lim_{x \rightarrow 0} \frac{f(x)}{x}$  equal to?

- (a) -1 (b) 0  
(c) 1 (d) 2

[NDA – 2023 (1)]

61. What is  $\lim_{x \rightarrow 0} \frac{f(x)}{x^2}$  equal to?

- (a) -1 (b) 0  
(c) 1 (d) 2

[NDA – 2023 (1)]

Consider the following for the next (02) items that follow

$$\text{Let } f(x) = \frac{a^{x-1} + b^{x-1}}{2} \text{ and } g(x) = x-1$$

62. What is  $\lim_{x \rightarrow 1} \frac{f(x)-1}{g(x)}$  equal to?

- (a)  $\frac{\ell n(ab)}{4}$  (b)  $\frac{\ell n(ab)}{2}$   
(c)  $\ln(ab)$  (d)  $2\ln(ab)$

[NDA-2023 (2)]

63. What is  $\lim_{x \rightarrow 1} f(x)^{\frac{1}{g(x)}}$  equal to?

- (a)  $\sqrt{ab}$  (b)  $ab$   
(c)  $2ab$  (d)  $\frac{\sqrt{ab}}{2}$

[NDA-2023 (2)]

Consider the following for the next (02) items that follow

Let  $f(x) = |x|$  and  $g(x) = [x] - 1$ , where  $[.]$  is the greatest integer function.

$$\text{Let } h(x) = \frac{f(g(x))}{g(f(x))}$$

64. What is  $\lim_{x \rightarrow 0+} h(x)$  equal to?

- (a) -2 (b) -1  
(c) 0 (d) 1

[NDA-2023 (2)]

65. What is  $\lim_{x \rightarrow 0-} h(x)$  equal to?

- (a) -2 (b) -1  
(c) 0 (d) 2

[NDA-2023 (2)]

66. Let  $f(x) = |x| + 1$  and  $g(x) = [x] - 1$ . Where  $[.]$  is the greatest integer function ,

$$\text{Let } h(x) = \frac{f(x)}{g(x)}$$

What is  $\lim_{x \rightarrow 0-} h(x) + \lim_{x \rightarrow 0+} h(x)$  equal to?

- (a)  $-\frac{3}{2}$  (b)  $-\frac{1}{2}$   
(c)  $\frac{1}{2}$  (d)  $\frac{3}{2}$

[NDA-2024 (1)]

67. Which one of the following is correct regarding  $\lim_{x \rightarrow 3} \frac{|x-3|}{x-3}$ ?

- (a) Limit exists and is equal to 1  
(b) Limit exists and is equal to 0  
(c) Limit exists and is equal to -1  
(d) Limit does not exist

[NDA-2024 (2)]

68. What is  $\lim_{\theta \rightarrow \frac{\pi}{2}} (\sec \theta - \tan \theta)$  equal to ?

- (a) -1 (b) 0  
(c)  $\frac{1}{2}$  (d) 1

[NDA-2024 (2)]

Consider the following for the two (02) items that follow:

Let the function  $f(x) = \sin[x]$ , where  $[.]$  is the greatest integer function and  $g(x) = |x|$

69. What is  $\lim_{x \rightarrow 0} \{f(x)g(x)\}$  equal to?

- (a) -1 (b) 0  
(c) 1 (d) Limit does not exist

[NDA-2025 (1)]

70. What is  $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$  equal to?

- (a)  $-\sin 1$  (b)  $\sin 1$   
(c) 0 (d) Limit does not exist

[NDA-2025 (1)]

Consider the following for the one (01) items that follow:

Let the function  $f(x) = x^2 - 1$

71. What is  $\lim_{x \rightarrow 1} \{f \circ f(x)\}$  equal to?

- (a) -1 (b) 0  
(c) 1 (d) 2

[NDA-2025 (1)]

Consider the following for the one (01) items that follow:

Let the function  $f(x) = x^2 + 9$

72. What is  $\lim_{x \rightarrow 0} \frac{\sqrt{f(x)} - 3}{\sqrt{f(x)} + 7 - 4}$  equal to?

- (a)  $\frac{2}{3}$  (b) 1  
(c)  $\frac{4}{3}$  (d) 2

[NDA-2025 (1)]

For the following two (02) items:  
Let

$$f(x) = \begin{cases} \frac{1 - \cos 2x}{x^2} & x < 0 \\ 9 & x = 0 \\ \sqrt{\sqrt{16 + \sqrt{x}} - 4} & x > 0 \end{cases}$$

73. What is  $\lim_{x \rightarrow 0^+} f(x)$  equal to?
- (a) 2 (b) 4  
(c) 6 (d) 8

[NDA-2025 (2)]

74. What is  $\lim_{x \rightarrow 0^+} f(x)$  equal to?
- (a) 6 (b) 7  
(c) 8 (d) 9

[NDA-2025 (2)]

For the following **one** items:

Consider the function  $f(x) = x|x|$

75. What is  $\lim_{x \rightarrow -1} f(x)$  equal to?
- (a) -1 (b) 0  
(c) 1 (d) Limit does not exist

[NDA-2025 (2)]

## ANSWER KEY

|     |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |     |   |
|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|-----|---|
| 1.  | a | 2.  | a | 3.  | c | 4.  | d | 5.  | b | 6.  | b | 7.  | b | 8.  | c | 9.  | a | 10. | d |
| 11. | d | 12. | a | 13. | c | 14. | c | 15. | c | 16. | c | 17. | d | 18. | b | 19. | a | 20. | d |
| 21. | a | 22. | b | 23. | c | 24. | d | 25. | d | 26. | b | 27. | c | 28. | a | 29. | a | 30. | b |
| 31. | c | 32. | a | 33. | c | 34. | d | 35. | a | 36. | d | 37. | c | 38. | a | 39. | c | 40. | b |
| 41. | a | 42. | d | 43. | c | 44. | a | 45. | c | 46. | c | 47. | b | 48. | c | 49. | d | 50. | c |
| 51. | a | 52. | c | 53. | d | 54. | a | 55. | d | 56. | c | 57. | d | 58. | d | 59. | b | 60. | b |
| 61. | a | 62. | b | 63. | a | 64. | b | 65. | a | 66. | a | 67. | d | 68. | b | 69. | b | 70. | d |
| 71. | a | 72. | c | 73. | a | 74. | d | 75. | a |     |   |     |   |     |   |     |   |     |   |

## Solutions

**Sol. 1. (a)**

$$\lim_{x \rightarrow \infty} (\sqrt{a^2 x^2 + ax + 1} - \sqrt{a^2 x^2 + 1})$$

rationalize

$$\lim_{x \rightarrow \infty} (\sqrt{a^2 x^2 + ax + 1} - \sqrt{a^2 x^2 + 1}) \times \frac{(\sqrt{a^2 x^2 + ax + 1} + \sqrt{a^2 x^2 + 1})}{(\sqrt{a^2 x^2 + ax + 1} + \sqrt{a^2 x^2 + 1})}$$

$$\lim_{x \rightarrow \infty} \frac{(a^2 x^2 + ax + 1 - a^2 x^2 - 1)}{(\sqrt{a^2 x^2 + ax + 1} + \sqrt{a^2 x^2 + 1})}$$

$$\lim_{x \rightarrow \infty} \frac{(ax)}{(\sqrt{a^2 x^2 + ax + 1} + \sqrt{a^2 x^2 + 1})} = \frac{a}{a + a} = \frac{1}{2}$$

**Sol. 2. (a)**

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$$

$= 0 \times (\text{any value between } -1 \text{ and } 1) = 0$

**Sol. 3. (c)**

$$\lim_{x \rightarrow -2} \left( \frac{x+2}{x^3+8} \right)$$

apply L. Hospital

$$\lim_{x \rightarrow -2} \left( \frac{1}{3x^2} \right) = \frac{1}{12}$$

**Sol. 4. (d)**

$$\lim_{x \rightarrow 0} \frac{2(1 - \cos x)}{x^2}$$

apply L. Hospital

$$\lim_{x \rightarrow 0} \frac{2(\sin x)}{2x} = 1$$

**Sol. 5. (b)**

$$\lim_{x \rightarrow 2} \frac{x-2}{x^2-4}$$

apply L. Hospital

$$\lim_{x \rightarrow 2} \frac{1}{2x} = \frac{1}{4}$$

**Sol. 6. (b)**

$$\text{I. } \lim_{x \rightarrow 0} \frac{1}{x}$$

$$\text{R.H.L. } \lim_{h \rightarrow 0} \frac{1}{h} = \infty$$

$$\text{L.H.L. } \lim_{h \rightarrow 0} \frac{1}{-h} = -\infty$$

$$\lim_{x \rightarrow 0} \frac{1}{x} \text{ does not exist.}$$

II.  $\lim_{x \rightarrow 0} e^{1/x}$  does not exist

$$\text{R.H.L. } \lim_{h \rightarrow 0} e^{1/h} = \infty$$

$$\text{L.H.L. } \lim_{h \rightarrow 0} e^{1/-h} = 0$$

$$\lim_{x \rightarrow 0} \frac{1}{x} \text{ does not exist.}$$

only II statement is correct.

**Sol. 7. (b)**

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$$

apply L. Hospital

$$\lim_{x \rightarrow 0} \frac{\frac{1}{2\sqrt{1+x}}}{1} = \frac{1}{2}$$

**Sol. 8. (c)**

$$\lim_{x \rightarrow 0} \sin \frac{1}{x}$$

its value oscillates between -1 and 1 so limit does not exist.

$$\text{now } \lim_{x \rightarrow 0} x \sin \frac{1}{x}$$

$= 0 \times (\text{any value between } -1 \text{ and } 1) = 0$

limit exists.

**Sol. 9. (a)**

$$\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x} = 0$$

standard limits.

**Sol. 10. (d)**

$$\lim_{x \rightarrow 0} \frac{1 - \sqrt{1+x}}{x}$$

apply L. Hospital

$$\lim_{x \rightarrow 0} \frac{-\frac{1}{2\sqrt{1+x}}}{1} = -\frac{1}{2}$$

**Sol. 11. (d)**

$$\lim_{x \rightarrow 2} \frac{2-x}{x^3-8}$$

apply L. hospital

$$\lim_{x \rightarrow 2} \frac{-1}{3x^2} = \frac{-1}{12}$$

**Sol. 12. (a)**

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$$

apply L. hospital

$$\lim_{x \rightarrow 0} \frac{\sin x}{1} = 0$$

**Sol. 13. (c)**

$$\lim_{x \rightarrow 0} \frac{\cos x}{\pi - x} = \frac{1}{\pi}$$

**Sol. 14. (c)**

$$\lim_{x \rightarrow 0} \frac{\sin 2x + 4x}{2x + \sin 4x}$$

apply L. Hospital

$$\lim_{x \rightarrow 0} \frac{2 \cos 2x + 4}{2 + 4 \cos 4x} = \frac{6}{6} = 1$$

**Sol. 15. (c)**

$$\lim_{x \rightarrow 9} \frac{\sqrt{f(x)} - 3}{\sqrt{x} - 3}$$

apply L. hospital

$$\lim_{x \rightarrow 9} \frac{\frac{1}{2\sqrt{f(x)}} f'(x)}{\frac{1}{2\sqrt{x}}} = \lim_{x \rightarrow 9} \frac{\sqrt{x} \times f'(x)}{\sqrt{f(x)}}$$

$$= \frac{\sqrt{9} \times f'(9)}{\sqrt{f(9)}} = \frac{4 \times 3}{3} = 4$$

**Sol. 16. (c)**

$$\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x}$$

apply L. Hospital

$$\lim_{x \rightarrow 0} \frac{n(1+x)^{n-1}}{1} = n$$

**Sol. 17. (d)**

$$\lim_{x \rightarrow 0} \frac{x}{\sqrt{1 - \cos x}}$$

$$\lim_{x \rightarrow 0} \frac{x}{\sqrt{2} \left| \sin \frac{x}{2} \right|}$$

L.H.L is negative and R.H.L. is positive for this.  
So limit does not exist.

**Sol. 18. (c)**

$$\lim_{x \rightarrow 0} \frac{\log_5(1+x)}{x}$$

apply L. Hospital

$$\lim_{x \rightarrow 0} \frac{1}{1+x} \cdot \frac{1}{\log_5 e} = \log_e 5$$

**Sol. 19. (a)**

$$\lim_{x \rightarrow 0} \frac{5^x - 1}{x}$$

apply L. Hospital

$$\lim_{x \rightarrow 0} \frac{5^x \cdot \log_e 5}{1} = \log_e 5$$

**Sol. 20. (d)**

$$\lim_{n \rightarrow \infty} \frac{1+2+3+\dots+n}{1^2+2^2+3^2+\dots+n^2}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{n(n+1)}{2}}{\frac{n(n+1)(2n+1)}{6}} = \lim_{n \rightarrow \infty} \frac{3}{2n+1} = 0$$

**Sol. 21. (a)**

$$\lim_{x \rightarrow 1} \frac{G(x) - f(1)}{x - 1}$$

apply L. Hospital

$$\lim_{x \rightarrow 1} \frac{G(x) - f(1)}{x - 1} = G'(x)$$

$$\text{Now, } G'(x) = \frac{-2x}{2\sqrt{25-x^2}}$$

$$G'(1) = \frac{-2}{2\sqrt{25-1}} = \frac{-1}{\sqrt{24}} = \frac{-1}{2\sqrt{6}}$$

**Sol. 22. (b)**

$$\lim_{x \rightarrow \infty} \left( \frac{2+x^2}{1+x} - Ax - B \right) = 3$$

$$\lim_{x \rightarrow \infty} \left( \frac{2+x^2 - Ax(1+x) - B(1+x)}{1+x} \right) = 3$$

$$\lim_{x \rightarrow \infty} \left( \frac{x^2(1-A) - x(A+B) + 2-B}{1+x} \right) = 3$$

for definite result of this algebraic limit coefficient of  $x^2$  should be zero.

$$\Rightarrow 1-A = 0 \text{ so } A = 1$$

$$-(A+B) = 3 \text{ so } B = -4$$

**Sol. 23. (c)**

solution in above question.

**Sol. 24. (d)**

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$$

apply L. Hospital

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = f'(x)$$

$$\text{Now, } f'(x) = \frac{-2x}{2\sqrt{25-x^2}}$$

$$f'(1) = \frac{-2}{2\sqrt{25-1}} = \frac{-1}{\sqrt{24}}$$

**Sol. 25. (d)**

$$\lim_{x \rightarrow 2} \frac{\sin(e^{x-2} - 1)}{\ln(x-1)}$$

apply L. Hospital

$$\lim_{x \rightarrow 2} \frac{\cos(e^{x-2} - 1)e^{x-2}}{\frac{1}{x-1}} = 1$$

**Sol. 26. (b)**

$$\text{Given } f(x) = \frac{a^{[x]+x} - 1}{[x] + x}$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{a^{[x]+x} - 1}{[x] + x} = \lim_{h \rightarrow 0} \frac{a^{[0+h]+(0+h)} - 1}{[0+h] + (0+h)}$$

$$\lim_{h \rightarrow 0} \frac{a^{[h]+(h)} - 1}{[h] + (h)} = \lim_{h \rightarrow 0} \frac{a^{(h)} - 1}{(h)} = \log_e a$$

**Sol. 27. (c)**

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{a^{[x]+x} - 1}{[x] + x} = \lim_{h \rightarrow 0} \frac{a^{[0-h]+(0-h)} - 1}{[0-h] + (0-h)}$$

$$\lim_{h \rightarrow 0} \frac{a^{-1+(h)} - 1}{-1 + (h)} = \frac{a^{-1} - 1}{-1} = 1 - a^{-1}$$

**Sol. 28. (a)**

$$\lim_{x \rightarrow 0} e^{-\frac{1}{x^2}} = e^{-\frac{1}{0}} = 0$$

**Sol. 29. (a)**

$$\lim_{x \rightarrow 0} \phi(x) = a^2$$

$$\lim_{x \rightarrow 0} \phi\left(\frac{x}{a}\right) = a^2$$

because function value is constant.

**Sol. 30. (b)**

$$\lim_{x \rightarrow 0} \frac{e^x - (1+x)}{x^2}$$

apply L. hospital rule

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{2x}$$

again apply L. hospital rule

$$\lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$$

**Sol. 31. (c)**

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$$

apply L. Hospital

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = f'(x)$$

$$\text{Now, } f'(x) = \frac{-2x}{2\sqrt{9-x^2}}$$

$$f'(1) = \frac{-2}{2\sqrt{9-1}} = \frac{-1}{2\sqrt{2}}$$

**Sol. 32. (a)**

If  $\lim_{x \rightarrow a} f(x)$  and  $\lim_{x \rightarrow a} g(x)$  both exists, then

$$\lim_{x \rightarrow a} \{f(x)g(x)\} \text{ exists. But if } \lim_{x \rightarrow a} \{f(x)g(x)\}$$

exists, then it is not necessary that both  $\lim_{x \rightarrow a} f(x)$

and  $\lim_{x \rightarrow a} g(x)$  exists.

**Sol. 33. (c)**

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{x} = l \text{ and } \lim_{x \rightarrow \infty} \frac{\cos x}{x} = m,$$

$$l = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{x} = \frac{1}{\frac{\pi}{2}} = \frac{2}{\pi}$$

$$m = \lim_{x \rightarrow \infty} \frac{\cos x}{x} = \frac{\text{value between } -1 \text{ to } 1}{\infty} = 0$$

**Sol. 34. (d)**

$$\lim_{h \rightarrow 0} \frac{\sqrt{2x+3h} - \sqrt{2x}}{2h}$$

rationalize the numerator

$$\lim_{h \rightarrow 0} \frac{\sqrt{2x+3h} - \sqrt{2x}}{2h} \times \frac{\sqrt{2x+3h} + \sqrt{2x}}{\sqrt{2x+3h} + \sqrt{2x}}$$

$$\lim_{h \rightarrow 0} \frac{2x+3h-2x}{2h(\sqrt{2x+3h} + \sqrt{2x})} = \frac{3}{4\sqrt{2x}}$$

**Sol. 35. (a)**

$$\lim_{x \rightarrow 0} \frac{\tan x}{\sin 2x} \times \frac{2x}{2x}$$

$$= \frac{\tan x}{x} \times \frac{2x}{\sin 2x} \times \frac{1}{2} = \frac{1}{2}$$

**Sol. 36. (d)**

$$\lim_{x \rightarrow \frac{\pi}{6}} \frac{2\sin^2 x + \sin x - 1}{2\sin^2 x - 3\sin x + 1}$$

$$\lim_{x \rightarrow \frac{\pi}{6}} \frac{(\sin x + 1)(2\sin x - 1)}{(\sin x - 1)(2\sin x - 1)} = \lim_{x \rightarrow \frac{\pi}{6}} \frac{(\sin x + 1)}{(\sin x - 1)} = -3$$

**Sol. 37. (c)**

$$\lim_{\theta \rightarrow 0} \frac{\sqrt{1-\cos \theta}}{\theta} = \lim_{\theta \rightarrow 0} \frac{\sqrt{2} \sin(\theta/2)}{\theta}$$

$$= \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

**Sol. 38. (a)**

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$$

apply L. Hospital

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = f'(x)$$

$$\text{Now, } f'(x) = \frac{-2x}{2\sqrt{25-x^2}}$$

$$f'(1) = \frac{-2}{2\sqrt{25-1}} = \frac{-1}{\sqrt{24}}$$

**Sol. 39. (c)**

$$\lim_{x \rightarrow 0} \frac{1 - \cos^3 4x}{x^2}$$

apply L. Hospital rule

$$\lim_{x \rightarrow 0} \frac{3\cos^2 4x \cdot \sin 4x \cdot 4}{2x}$$

$$\lim_{x \rightarrow 0} \frac{3\cos^2 4x \cdot \sin 4x \cdot 4}{4x} \times 2$$

$$\left( \because \lim_{x \rightarrow 0} \frac{\sin 4x}{4x} = 1 \right)$$

$$= 3(4)(2) = 24$$

**Sol. 40. (b)**

$$\lim_{x \rightarrow 0} \frac{\sin x^0}{\tan 3x^0} \times \frac{3x^0}{3x^0}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x^0}{x^0} \times \frac{3x^0}{\tan 3x^0} \times \frac{1}{3} = \frac{1}{3}$$

**Sol. 41. (a)**

$$\lim_{x \rightarrow 0} \frac{3^x + 3^{-x} - 2}{x}$$

apply L. hospital

$$\lim_{x \rightarrow 0} \frac{3^x \log 3 - 3^{-x} \log 3}{1} = 0$$

**Sol. 42. (d)**

$$\lim_{x \rightarrow 1} \frac{x + x^2 + x^3 - 3}{x - 1}$$

apply L. hospital

$$\lim_{x \rightarrow 1} \frac{1 + 2x + 3x^2}{1} = 6$$

**Sol. 43. (c)**

$$\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{x^3 - k^3}{x^2 - k^2}$$

apply L. hospital

$$\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{4x^3}{1} = 4$$

and

$$\lim_{x \rightarrow k} \frac{x^3 - k^3}{x^2 - k^2} = \lim_{x \rightarrow k} \frac{3x^2}{2x} = \lim_{x \rightarrow k} \frac{3x}{2} = \frac{3k}{2}$$

$$\frac{3k}{2} = 4 \Rightarrow k = \frac{8}{3}$$

**Sol. 44. (a)**

$$\lim_{x \rightarrow 0} \frac{\sin x \log(1 - x)}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{\log(1 - x)}{x} = 1(-1) = -1$$

**Sol. 45. (c)**

$$\lim_{x \rightarrow -1} \frac{f(x)}{x^2 - 1} = -\frac{3}{2}$$

by applying limit it gives zero in denominator but answer is a real number given so numerator will be also zero by applying limit.

$$\lim_{x \rightarrow -1} f(x) = f(-1) = 0$$

**Sol. 46. (c)**

$$\lim_{x \rightarrow a} \frac{a^x - x^a}{x^a - a^a} = -1$$

apply L. hospital

$$\lim_{x \rightarrow a} \frac{a^x \log a - ax^{a-1}}{ax^{a-1} - a^a} = -1$$

$$\frac{a^a \log a - a \cdot a^{a-1}}{a \cdot a^{a-1}} = -1$$

$$\frac{a^a \log a - a}{a} = -1$$

$$a^a \log a - a = -a$$

$$a^a \log a = 0$$

$$a = 1$$

**Sol. 47. (b)**

$$\lim_{x \rightarrow -1} \frac{x^3 + x^2}{x^2 + 3x + 2}$$

apply L. hospital

$$\lim_{x \rightarrow -1} \frac{3x^2 + 2x}{2x + 3} = \frac{3 - 2}{-2 + 3} = 1$$

**Sol. 48. (c)**

$$\lim_{n \rightarrow \infty} \frac{a^n + b^n}{a^n - b^n} \quad a > b > 1 \text{ as } a > b$$

$$\Rightarrow \frac{b}{a} < 1$$

$$\text{So } \lim_{n \rightarrow \infty} \frac{a^n \left[ 1 + \left( \frac{b}{a} \right)^n \right]}{a^n \left[ 1 - \left( \frac{b}{a} \right)^n \right]}$$

As  $n \rightarrow \infty$  and  $\frac{b}{a} < 1$  So  $\left( \frac{b}{a} \right)^n = 0$

$$\text{So } \lim_{n \rightarrow \infty} \frac{1 + \left( \frac{b}{a} \right)^n}{1 - \left( \frac{b}{a} \right)^n}$$

$$= \frac{1}{1} = 1$$

**Sol. 49. (d)**

$$f(x) = \begin{cases} 1 + \frac{x}{2k}; & 0 < x < 2 \\ kx; & 2 < x < 4 \end{cases}$$

given  $\lim_{n \rightarrow 2} f(x)$

as  $\lim_{n \rightarrow 2} f(x)$  exist

so  $\lim_{n \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} f(x), \quad 1 + \frac{2}{2k} = 2k$

$$1 + \frac{1}{k} = 2k$$

$$K + 1 = 2k^2$$

$$2k^2 - k - 1 = 0$$

$$2k^2 - 2k + k - 1 = 0$$

$$2k(k-1) + (k-1) = 0$$

$$(k-1)(2k+1) = 0$$

$$K = 1 \text{ or } k = -\frac{1}{2}$$

**Sol. 50. (c)**

$$f(x) = \begin{cases} \frac{[x]}{|x|} & x \neq 0 \end{cases}$$

$$f(x) = \begin{cases} 0 & x < 1 \\ 1 & x = 1 \\ \frac{1}{x} & x > 1 \end{cases}$$

$$\lim_{x \rightarrow 1^+} \frac{[x]}{|x|} [1 + h] = 1$$

Or  $\lim_{h \rightarrow 0} \frac{f(1+h)}{1+h}$

$$\lim_{h \rightarrow 0} \frac{[1+h]}{1+h} = \frac{1}{1} = 1$$

**Sol. 51. (a)**

$$\lim_{x \rightarrow 0} x^3 (\operatorname{cosec} x)^2$$

$$= \lim_{x \rightarrow 0} \frac{x^3}{\sin^2 x}$$

$$= \lim_{x \rightarrow 0} x \cdot \frac{x^2}{\sin^2 x}$$

$$= 0 \times 1 = 0$$

**Sol. 52. (c)**

$$= \lim_{x \rightarrow 1} \frac{x^3 - 1}{\sqrt{x} - 1}$$

Using L.hospital rule

$$\Rightarrow \lim_{x \rightarrow 1} \frac{3x^2}{1/2\sqrt{x}}$$

$$\Rightarrow \lim_{x \rightarrow 1} 6x^2 \sqrt{x} = 6$$

**Sol. 53. (d)**

$$\lim_{x \rightarrow 0} \frac{x}{\sqrt{1 - \cos 4x}}$$

$$= \lim_{x \rightarrow 0} \frac{x}{\sqrt{2 \sin^2 2x}} = \lim_{x \rightarrow 0} \frac{2x}{2\sqrt{2} |\sin 2x|}$$

For R.H.L.

$$= \lim_{h \rightarrow 0} \frac{2h}{2\sqrt{2} \sin 2h} = \frac{1}{2\sqrt{2}}$$

For L.H.L.

$$= \lim_{h \rightarrow 0} \frac{-2h}{2\sqrt{2} \sin 2h} = -\frac{1}{2\sqrt{2}}$$

Limit does not exist.

**Sol. 54. (a)**

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{4x - 2\pi}{\cos x}$$

Apply L hospital rule

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{4}{-\sin x} = -4$$

**Sol. 55. (d)**

$$f(x) = \frac{x^2 + x + |x|}{x}$$

R.H.L. at  $x = 0$

$$= \lim_{h \rightarrow 0} \frac{h^2 + h + h}{h} = \frac{h^2 + 2h}{h} = h + 2 = 2$$

L.H.L. at  $x = 0$

$$= \lim_{h \rightarrow 0} \frac{h^2 + h - h}{h} = \frac{h^2}{h} = h = 0$$

$\lim_{x \rightarrow 0} f(x)$  does not exist

**Sol. 56. (c)**

$$\lim_{h \rightarrow 0} \frac{\sin^2(x+h) - \sin^2 x}{h}$$

Apply L hospital rule

Diff. w.r.t. h

$$\lim_{h \rightarrow 0} \frac{2 \sin(x+h) \cos(x+h)}{1}$$

$$= 2 \sin x \cos x = \sin 2x$$

**Sol. 57. (d)**

$$\lim_{x \rightarrow 5} \frac{5 - x}{|x - 5|}$$

R.H.L. at  $x = 5$

$$\lim_{h \rightarrow 0} \frac{5 - (5+h)}{|5+h-5|} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1$$

L.H.L. at  $x = 5$

$$\lim_{h \rightarrow 0} \frac{5 - (5-h)}{|5-h-5|} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$$

Limit does not exist.

**Sol. 58. (d)**

$$\lim_{x \rightarrow 1} \frac{x^9 - 1}{x^3 - 1}$$

For R.H.L

$$\lim_{x \rightarrow 1} \frac{x^9 - 1}{x^3 - 1}$$

Apply L. hospital rule

$$\lim_{x \rightarrow 1} \frac{9x^8}{3x^2} = 3$$

For L.H.L

$$\lim_{x \rightarrow 1} \frac{x^9 - 1}{(x^3 - 1)}$$

Apply L. hospital rule

$$\lim_{x \rightarrow 1} \frac{9x^8}{3x^2} = -3$$

Limit does not exists

**Sol. 59. (b)**

$$f(x) = \begin{vmatrix} \cos x & x & 1 \\ 2 \sin x & x^2 & 2x \\ \tan x & x & 1 \end{vmatrix}$$

$$f(0) = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 0$$

**Sol. 60. (b)**

$$\frac{f(x)}{x} = \begin{vmatrix} \cos x & 1 & 1 \\ 2 \sin x & x & 2x \\ \tan x & 1 & 1 \end{vmatrix}$$

$$= \lim_{x \rightarrow 0} \frac{f(x)}{x} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 0$$

**Sol. 61. (a)**

$$\frac{f(x)}{x^2} = \begin{vmatrix} \cos x & 1 & 1 \\ \sin x & 1 & 2 \\ \tan x & 1 & 1 \end{vmatrix}$$

$$= \lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 0 & 1 & 1 \end{vmatrix} = -1$$

**Sol. 62. (b)**

$$\lim_{x \rightarrow 1} \frac{a^{x-1} + b^{x-1} - 1}{2(x-1)}$$

Apply L. hospital rule

$$= \lim_{x \rightarrow 1} \frac{a^{x-1} \log a + b^{x-1} \log b}{2} = \frac{\log a + \log b}{2} = \frac{1}{2} \log ab$$

**Sol. 63. (a)**

$$\lim_{x \rightarrow 1} f(x)^{\frac{1}{g(x)}}$$

$$y = \lim_{x \rightarrow 1} \left( \frac{a^{x-1} + b^{x-1}}{2} \right)^{\frac{1}{x-1}}$$

taking log both sides

$$\log y = \lim_{x \rightarrow 1} \frac{\log \left( \frac{a^{x-1} + b^{x-1}}{2} \right)}{x-1}$$

Apply L. hospital rule

$$\log y = \lim_{x \rightarrow 1} \frac{\left( \frac{2}{a^{x-1} + b^{x-1}} \right) \left( \frac{a^{x-1} \log a + b^{x-1} \log b}{2} \right)}{1}$$

$$\log y = \frac{\left( \frac{2}{1+1} \right) \left( \frac{\log a + \log b}{2} \right)}{1} = \frac{1}{2} \log ab$$

$$\log y = \log \sqrt{ab}$$

$$y = \sqrt{ab}$$

**Sol. 64. (b)**

$$h(x) = \frac{f(g(x))}{g(f(x))} = \frac{|[x]-1|}{|[x]-1|}$$

$$\lim_{x \rightarrow 0+} h(x) = \text{R.H.L at } x = 0$$

$$\lim_{h \rightarrow 0} \frac{|[h]-1|}{|[h]-1|} = \lim_{h \rightarrow 0} \frac{|0-1|}{[h]-1} = \frac{1}{0-1} = -1$$

**Sol. 65. (a)**

$$h(x) = \frac{f(g(x))}{g(f(x))} = \frac{|[x]-1|}{|[x]-1|}$$

$$\lim_{x \rightarrow 0+} h(x) = \text{L.H.L at } x = 0$$

$$\lim_{h \rightarrow 0} \frac{|[-h]-1|}{|[-h]-1|} = \lim_{h \rightarrow 0} \frac{|-1-1|}{[h]-1} = \frac{2}{0-1} = -2$$

**Sol. 66. (a)**

$$f(x) = |x| + 1$$

$$g(x) = [x] - 1$$

$$h(x) = \frac{f(x)}{g(x)}$$

$$\lim_{x \rightarrow 0-} h(x) + \lim_{x \rightarrow 0+} h(x)$$

$$= \lim_{x \rightarrow 0-} \frac{|x|+1}{[x]-1} + \lim_{x \rightarrow 0+} \frac{|x|+1}{[x]-1}$$

$$= \lim_{h \rightarrow 0} \frac{|0-h|+1}{[0-h]-1} + \lim_{h \rightarrow 0} \frac{|0+h|+1}{[0+h]-1}$$

$$= \lim_{h \rightarrow 0} \frac{h+1}{-1-1} + \lim_{h \rightarrow 0} \frac{h+1}{0-1}$$

$$= \frac{1}{-2} + \frac{1}{-1} = -\frac{3}{2}$$

**Sol. 67. (d)**

$$\lim_{x \rightarrow 3} \frac{x-3}{x-3}$$

R.H.L. at  $x = 3$

$$\lim_{h \rightarrow 0} \frac{3+h-3}{3+h-3} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$$

L.H.L. at  $x = 3$

$$\lim_{h \rightarrow 0} \frac{3-h-3}{3-h-3} = \lim_{h \rightarrow 0} \frac{h}{-h} = -1$$

Limit does not exist.

**Sol. 68. (b)**

$$\lim_{\theta \rightarrow \frac{\pi}{2}} (\sec \theta - \tan \theta)$$

$$= \lim_{\theta \rightarrow \frac{\pi}{2}} \left( \frac{1 - \sin \theta}{\cos \theta} \right)$$

apply L. Hospital rule

$$= \lim_{\theta \rightarrow \frac{\pi}{2}} \left( \frac{-\cos \theta}{-\sin \theta} \right) = 0$$

**Sol. 69. (b)**

$$\lim_{x \rightarrow 0} \{f(x)g(x)\}$$

For R.H.L.

$$\lim_{h \rightarrow 0} \sin[h] |h| = \lim_{h \rightarrow 0} \sin 0 |h| = 0$$

For L.H.L.

$$\lim_{h \rightarrow 0} \sin[-h] |-h| = \lim_{h \rightarrow 0} \sin(-1) |-h| = 0$$

So limit is 0.

**Sol. 70. (d)**

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$$

For R.H.L.

$$\lim_{h \rightarrow 0} \frac{\sin[h]}{|h|} = \lim_{h \rightarrow 0} \frac{\sin 0}{|h|} = 0$$

For L.H.L.

$$\lim_{h \rightarrow 0} \frac{\sin[-h]}{|-h|} = \lim_{h \rightarrow 0} \frac{\sin(-1)}{|-h|} = \lim_{h \rightarrow 0} \frac{-\sin(1)}{h} = -\infty$$

limit does not exists.

**Sol. 71. (a)**

$$f(x) = x^2 - 1$$

$$fof(x) = (x^2 - 1)^2 - 1$$

$$fof(1) = (1 - 1)^2 - 1 = -1$$

**Sol. 72. (c)**

$$f(x) = x^2 + 9$$

$$f(0) = 9$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{f(x)} - 3}{\sqrt{f(x)} + 7 - 4}$$

Apply L hospital

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2\sqrt{f(x)}} f'(x)}{\frac{1}{2\sqrt{f(x)} + 7} f'(x)} = \frac{\frac{1}{2\sqrt{f(0)}}}{\frac{1}{2\sqrt{f(0)} + 7}} = \frac{\frac{1}{\sqrt{9}}}{\frac{1}{\sqrt{16}}} = \frac{4}{3}$$

**Sol. 73. (a)**

L.H.L. at 0

$$\lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{1 - \cos 2(-h)}{(-h)^2}$$

$$\lim_{h \rightarrow 0} \frac{1 - \cos 2(h)}{(h)^2} = \lim_{h \rightarrow 0} \frac{2 \sin^2 h}{h^2} = 2$$

**Sol. 74. (d)**

R.H.L. at 0

$$\lim_{h \rightarrow 0} f(0+h)$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{h}}{\sqrt{(16+\sqrt{h})}-4} \times \frac{\sqrt{(16+\sqrt{h})}+4}{\sqrt{(16+\sqrt{h})}+4}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{h}}{\sqrt{16+h}-16} \times \frac{\sqrt{(16+\sqrt{h})}+4}{1} = 8$$

**Sol. 75. (a)**

Left hand limit at  $x = -1$

$$\lim_{h \rightarrow 0} f(-1-h) = \lim_{h \rightarrow 0} (-1-h) |-1-h| = (-1) |-1| = -1$$